

RESEARCH ARTICLE

Evaluation of forecasted wind speed at turbine hub height and wind ramps by five NWP models with observations from 262 wind farms over China

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Abstract

Accurate wind speed forecasts are essential for optimizing the efficiency of wind energy operations. Currently, there is limited research on nationwide assessment of predictive performance in multiple numerical weather prediction (NWP) models for wind speed at turbine hub height over China, especially concerning wind ramp events. Utilizing observed measurements from 262 wind farms, this study evaluated the performance of five NWP models in forecasting the mean state and spatiotemporal variations of wind speed as well as wind ramps. The results indicated that the European Center for Medium-Range Weather Forecast Integrated Forecasting System (ECMWF-IFS) performed the best in forecasting climatological wind speed with a temporal correlation coefficient (TCC) of 0.74 and root mean square error (RMSE) of 2.34 m s^{-1} . Although not widely utilized in China, the model from Meteo-France (MF-ARPEGE) showed promising potential for wind energy forecasting with a TCC of 0.72 and RMSE of 2.45 m s^{-1} . In terms of temporal variations of wind speed, all the models could reasonably predict the seasonal variations of wind speed, whereas only three NWP models were able to capture the characteristics of the observed diurnal variation. An error decomposition analysis further revealed that the primary source of predicted error for wind speed was the sequence error component (SEQU), indicating the model errors were mainly attributed from the temporal inconsistency between forecasts and observations. Furthermore, the occurrences of wind ramps were generally underestimated by NWP models, while this shortcoming can be partly overcome by improving the time resolution of NWP models.

KEYWORDS

error decomposition, NWP models, wind ramp event, wind speed at turbine hub-height

1 | INTRODUCTION

With the rapid growth of global economies, the Earth has experienced a dramatic warming due to increasing emissions of carbon dioxide and other pollutants (Nonki et al., 2023; Xue et al., 2021). China has made commitments to reach a carbon peak before 2030 and achieve carbon neutrality before 2060 (Song et al., 2022). To achieve the goals, the transition from fossil fuels to renewable clean energies is increasingly imperative. Wind energy generation is one of the key renewable energies and has experienced rapid growth in recent years (Li et al., 2022). However, wind energy generation is significantly dependent on the wind speed, owing to their cubic relationship within the range of 3 to 12 m s⁻¹. And even a small change of wind speed in that range could induce massive impact on energy output. For example, a 1 m s⁻¹ change in rotor-layer wind speeds from 7 to 8 m s⁻¹ can result in large energy output changes up to 50% (International Electrotechnical Commission, 2007).

The variability of wind speed presents significant challenges for wind power generation. Changes in wind speeds during short time periods such as a few hours could induce rapid, large changes in wind energy generation, known as wind ramps (Gallego-Castillo et al., 2015). With increased wind power penetration, the fluctuations of wind energy generated by wind ramps bring more challenges to the accuracy of real-time wind energy forecasting. The wind ramp could cause a sudden shift to bring other generation units on-line in a relatively short time to maintain the grid in balance, which can be costly (Djalalova et al., 2020). Therefore, the accuracy of wind forecasts and wind ramp events is crucial for the efficiency of wind energy operations and the integration of wind energy into electricity grids and markets (Olson et al., 2019).

In recent decades, numerical weather prediction (NWP) models have been developed and widely used in weather forecasting. Many previous studies have verified wind forecasts in NWP models using various instruments (Deng et al., 2023; Martín et al., 2014). For example, the forecasting performance of 10-m wind speed in NWP models in the context of wind forecasting was examined over northwestern China (Liu et al., 2021) and eastern China (Lei et al., 2023) based on in situ observations. Jonassen et al. (2021) evaluated the vertical profiles of wind speed in three NWP models over the Weddell Sea region with radio soundings.

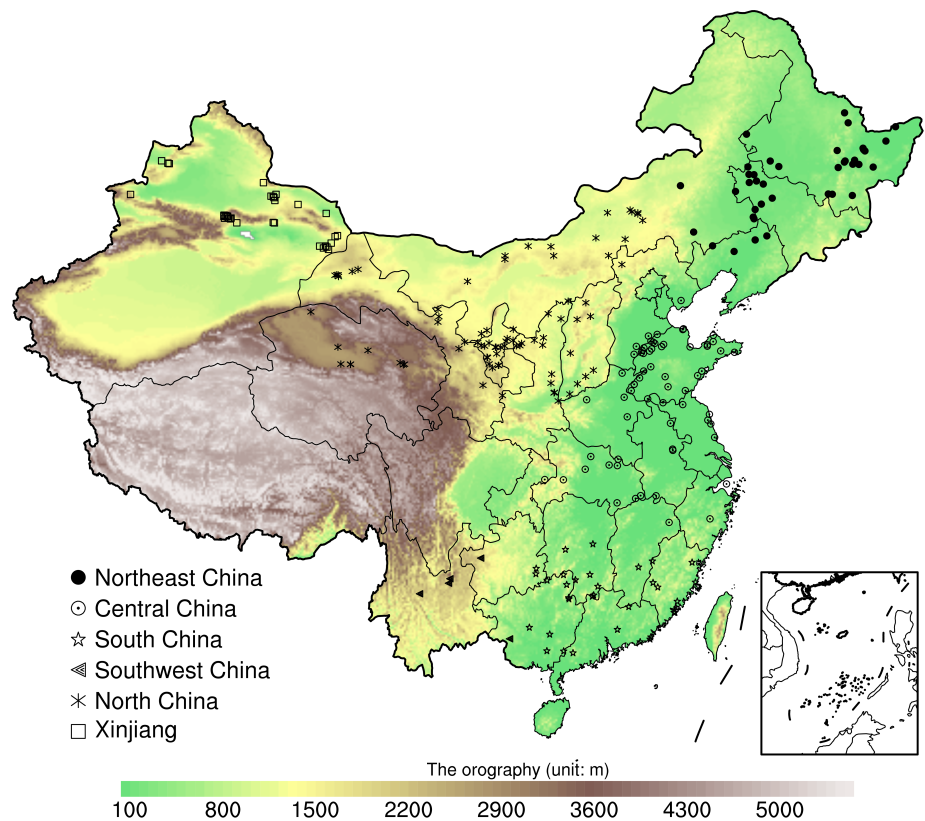
Wind turbines are usually operated at a height of 50 to 200 m above ground level (AGL), which is mostly within the surface layer and atmospheric boundary layer (He et al., 2014; Olson et al., 2019). The hub-height wind

speed is significantly influenced by the surface roughness, friction velocity and atmospheric stability (Gryning et al., 2007; Veers et al., 2019). Moreover, the diurnal evolution of the planetary boundary layer makes it a great challenge for NWP models to accurately forecast the hub-height wind speed, particularly during the sunset and sunrise periods (Berg et al., 2021; Heppelmann et al., 2017; Hu et al., 2013). Thus, the evaluations of hub-height wind speed forecasted by NWP models and further understanding of the model error are worthy of more study. Most previous studies on the assessment of forecasted hub-height wind speed by NWP models are over local areas using limited observations such as tall towers, turbine hubs and Doppler lidars. Chen et al. (2021) assessed wind speed performance at hub height in five wind parks forecasted by Norwegian NWP. Kalverla et al. (2018) evaluated the representation of wind and other relevant variables in three mainstream meteorological models based on unique met masts and lidars.

A wind ramp is identified when a large change (ranging from 20% to 50% of total capacity) in wind generation within a short period (ranging from 1 to 4 h) (Deppe et al., 2013; Lee et al., 2022). Though the term wind ramp event is widely used, there is no commonly accepted definition, particularly for the magnitude and duration. The number of ramp events is very sensitive to the threshold chosen. Bianco et al. (2016) developed a wind energy ramp tool and metric (RT&M) for evaluating the skill of NWP, which accounts for phase, duration and amplitude errors in the forecast and differentiates between the impacts of up and down ramp events. The forecasts of wind ramps were also demonstrated in previous studies. Ferreira et al. (2011) presented a survey on wind power ramp forecasting, and Gallego-Castillo et al. (2015) provided a review on the then-recent history of wind power ramp forecasting. The sensitivity of wind speed forecasts during wind ramp events were investigated, including initial conditions, horizontal grid spacing, type of event and parameters in the boundary layer scheme (Bianco et al., 2019; Cui et al., 2017). Zhang et al. (2017) reported that using short-term wind power forecasts helps to improve the accuracy of wind power ramp forecasting during the first Wind Forecast Improvement Project (WFIP).

The purpose of this study is to provide a systematic assessment of five NWP models' ability to predict the climatological characteristics as well as seasonal and diurnal variations of hub-height wind speed in China by utilizing observations of hub-height wind speed from 262 wind farms during 1 year. Through a bias-attributing analysis method, the model bias of hub-height wind speed in NWP models was examined for better understanding the model error. The wind ramp events forecast

FIGURE 1 Spatial distribution of 262 wind farms. Locations over Northeast China, Central China, South China, Southwest China, North China and Xinjiang are marked by the dot, hollow circle, pentagon, triangle, star and square, respectively. Shaded areas indicate the orography (unit: M) above sea level.



by NWP models were investigated further by utilizing a RT&M based on multiple definitions of wind ramps. These results are expected to provide valuable information to model users and model developers for further improvement.

The remainder of this paper is organized as follows: Section 2 describes the dataset used in this study, including observations and NWP models. Section 3 introduces the methodology, including evaluation metric, the bias, distribution, sequence error decomposition method and RT&M. Section 4 provides the reader with our results, and the conclusion and discussion are presented in Section 5.

2 | DATA

2.1 | Observational dataset

This study utilizes wind speed at turbine hub height (approximately 100 m AGL) measured by either cup anemometers or sonic anemometers installed on tower collected from 262 wind farms in China. The representativeness of each wind measurement point has been carefully and comprehensively considered to reflect the wind energy resources (Liu et al., 2023). The

abnormal data including constant and singular values are cleaned to improve the data quality. The domain is divided into six regions based on orography and climate zones, including Northeast China, Central China, South China, Southwest China, North China and Xinjiang, as shown in Figure 1. The dataset covers the period from 1 February 2021 to 31 January 2022 with a temporal resolution of 15 minutes.

2.2 | NWP models

The forecasted wind speed at 100 m height from five operational global high-resolution NWP models was evaluated, including the European Center for Medium-Range Weather Forecast Integrated Forecast System (ECMWF-IFS), Meteo-France Action de Recherche Petite Echelle Grande Echelle (MF-ARPEGE), Deutscher Wetterdienst ICOSahedral Nonhydrostatic (DWD-ICON), Canadian Meteorological Center Global Deterministic Prediction System (CMC-GDPS) and National Center for Environmental Prediction Global Forecast System (NCEP-GFS). The ECMWF-IFS, MF-ARPEGE and CMC-GDPS models produce output every 3 hours, while the DWD-ICON and NCEP-GFS models provide output every hour. In terms of horizontal grid spacing, ECMWF-IFS and

DWD-ICON use a grid spacing of $0.125^\circ \times 0.125^\circ$, CMC-GDPS uses $0.15^\circ \times 0.15^\circ$ and both MF-ARPEGE and NCEP-GFS use $0.25^\circ \times 0.25^\circ$. These models were widely employed in weather forecasting services (Cheng & Yang, 2023; Chinta et al., 2023), particularly for wind power forecasting (Wang et al., 2022; Yang et al., 2022). Apart from their high spatial and temporal resolution, ECMWF-IFS is popular in China due to its high accuracy, while NCEP-GFS is favored for being freely accessible.

The model output of 100-m wind speed was horizontally interpolated to match the locations of the 262 site measurements using the inverse distance weighted interpolation method based on the four closest grid points and was temporally interpolated at the temporal resolution of 15 minutes to match the observations. The universal time coordinated (UTC) time was converted to Beijing time (BJT). We extracted the data starting from 12:00 UTC (20:00 BJT) 1 day-ahead and covering the period from 00:00 to 23:45 BJT on the second day with a total of 96 time steps for analysis. The missing data from observations and models were both excluded in the analysis to ensure the reliability of the study.

3 | METHODOLOGY

3.1 | Evaluation metrics

To quantitatively assess the forecast results of different NWP models, several metrics were employed, including the pattern/temporal correlation coefficient (PCC/TCC), root mean square error (RMSE) and mean bias error (MBE):

$$\text{PCC/TCC} = \frac{\sum_{i=1}^n (f_i - \bar{f})(o_i - \bar{o})}{\sqrt{\sum_{i=1}^n (f_i - \bar{f})^2} \sqrt{\sum_{i=1}^n (o_i - \bar{o})^2}} \quad (1)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (f_i - o_i)^2} \quad (2)$$

where n indicates the total number of samples. The term f_i and o_i represent the forecast and observation of sample i , respectively. The term $\bar{f}$ and $\bar{o}$ refer to the average of forecasts and observations, respectively. PCC represents the correlation coefficient between respective values of the same variables at corresponding locations on two different maps—forecast and observed values. On the other hand, TCC denotes the correlation coefficient between forecast and observed values at the same time.

3.2 | Error decomposition method

The errors of forecasted wind speed at 100 m AGL in NWP models were examined by utilizing an error decomposition method, which was proposed by Hodson et al. (2021) and widely adopted to diagnose error source of simulations for better understanding the NWP models (Lyu et al., 2022; Xiang et al., 2023). Through the decomposition of model bias, the model bias can be divided into three components, that is, the bias, distribution and the sequence error, representing the systematic, spatial and temporal components of bias, respectively.

The error decomposition method was conducted as follows: firstly, the mean square error (MSE), that is, the square of RMSE, at each grid was calculated by Equation (3):

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (f_i - o_i)^2 \quad (3)$$

According to the decomposing method introduced by Geman et al. (1992), the MSE can be broken down into a bias term and a variance term:

$$\begin{aligned} \text{MSE}(\epsilon) &= E(\epsilon)^2 + E((\epsilon^2) - E(\epsilon)^2) \\ &= \text{Bias}(\epsilon)^2 + \text{Var}(\epsilon) \end{aligned} \quad (4)$$

where ϵ represents the forecast error of the model as the difference between the forecast and observation, while $E(\epsilon)$ represents the mean of the forecast error which was equal to $\text{Bias}(\epsilon)$, $\text{Var}(\epsilon)$ represents the variance of the forecast error. The bias component quantifies the ability of the model to reproduce the average characteristics of the observations, and the variance component quantifies the extent to which the model reproduces the observed variability. The variance component can be further decomposed to obtain a deeper understanding of model performance (Hodson et al., 2021). The derivation begins by monotonically sorting the model predictions and observations, then decomposing the MSE of the result:

$$\omega = \text{sort}(f) - \text{sort}(o) \quad (5)$$

$$\text{MSE}(\omega) = \text{Bias}(\omega)^2 + \text{Var}(\omega) \quad (6)$$

where $\text{sort}(f)$ and $\text{sort}(o)$ represent the sorted observations and forecasts, respectively, and ω represents the forecast error after sorting. Given that altering the sequence of the data does not impact the bias, it remains consistent both before and after the sorting. Meanwhile, the sorted observations and forecasts share the same time series, and the variance at this point, $V(\omega)$,

describes the error caused by the data distribution $\text{Distribution}(\epsilon)$; thus, the Equations (7) and (8) can be obtained:

$$\text{Var}(\omega) = \text{Distribution}(\epsilon) \quad (7)$$

$$\text{MSE}(\omega) = \text{Bias}(\epsilon)^2 + \text{Distribution}(\epsilon) \quad (8)$$

Furthermore, the difference between $M(\epsilon)$ and $\text{MSE}(\omega)$ can be attributed to the time series variation, $\text{Sequence}(\epsilon)$; thus, the following equation can be obtained:

$$\begin{aligned} \text{MSE}(\epsilon) - \text{MSE}(\omega) &= \text{Var}(\epsilon) - \text{Var}(\omega) \\ &= \text{Sequence}(\epsilon) \end{aligned} \quad (9)$$

Thus, the overall decomposition formula of MSE can be represented as follows:

$$\begin{aligned} \text{MSE}(\epsilon) &= \text{Bias}(\epsilon)^2 + \text{Var}(\epsilon) \\ &= \text{Bias}(\epsilon)^2 + (\text{Var}(\epsilon) - \text{Var}(\omega)) + \text{Var}(\omega) \\ &= \text{Bias}(\epsilon)^2 + \text{Sequence}(\epsilon) + \text{Distribution}(\epsilon) \end{aligned} \quad (10)$$

where $\text{Bias}(\epsilon)^2$ is the bias component (BIAS), which represents the ability of the models to reproduce the average characteristics of the observations, $\text{Sequence}(\epsilon)$ is the sequence error component (SEQU), which represents the error due to the forecast being ahead of or lagging behind the observations. $\text{Distribution}(\epsilon)$ is the distribution error component (DIST), which represents the error due to the difference in data distribution between the forecasts and observations. In order to transfer the units of error components from $(\text{m s}^{-1})^2$ into m s^{-1} , both sides of the equation are divided by the RMSE of model during the same period. Thus, the error decomposition of the RMSE is conducted. A more detailed description of calculation can be found in Lyu et al. (2022) and Xiang et al. (2023).

3.3 | Wind ramp tool and metric

To evaluate the performance of simulated wind ramp events of NWP models, the observational and predictive wind power were both calculated based on wind speed at turbine hub height using a generic/normalized power curve, which was reported in Wilczak et al. (2019). The predictive skill of wind ramps in NWP models were assessed by utilization of the RT&M method developed during the first WFIP (Akish et al., 2019; Bianco

et al., 2016; Wilczak et al., 2015). The RT&M consisted of three components: in the first component, ramp events were identified using parameters such as central time of the ramp (C_t), change in power (Δp) and duration (Δt) in the time series of observed and forecasted power; in the second component, the forecasted and observed ramp events were matched during the overlapping period; and the third component calculated the skill score of the NWP models on an hourly basis, considering errors in central time, duration and change in power. The score was determined through a function involving the cube root of the product of the errors on C_t , Δp and Δt (Bianco et al., 2016). The standard thresholds for Δp were 30%, 40%, 50%, 60% and 70% of the rated normalized capacity, and thresholds for Δt were set at 60, 120 and 180 min. A total of 15 distinct definitions were formulated using various combinations of Δp and Δt . The selection of these thresholds contributes to a comprehensive exploration of forecasted wind ramps by NWP models.

Both up and down ramp events were considered in the evaluation of forecasting skill of NWP models. By matching observed and modeled ramps, eight scenarios were derived: up/up, null/up, down/up, up/null, down/null, up/down, null/down and down/down. Then the ramp skill score was calculated for all possible ramp definitions through the division of total score by the total number of matched and unmatched ramp events. Unmatched ramps had a null skill score, while matched ramps were assigned a skill score from -1 to $+1$ based on the errors in power change ($|\text{model } \Delta p - \text{obs } \Delta p|$), time duration ($|\text{model } \Delta t - \text{obs } \Delta t|$) and central time difference ($|\text{model } C_t - \text{obs } C_t|$) [details can be referred in Equations (1)–(6) of Bianco et al., 2016]. A positive score indicated the model had predictive skill, and the higher score denoted the higher predictive skill, while a negative score indicated the model was worse than random.

4 | RESULTS

4.1 | Mean state of wind speed

The spatial distribution of climatological wind speed at 100 m is illustrated in Figure 2. The observed high (low) wind speeds mainly located in northern (southern) China, which was consistent with previous studies (Li et al., 2022; Liu et al., 2019). The one day-ahead performances of forecasted 100-m wind speed were evaluated across all wind farms. All the five NWP models reproduced this pattern reasonably, with the averaged PCC ranging from 0.44 in CMC-GDPS to 0.58 in ECMWF-IFS (Table 1). These correlations were statistically significant at the 1% level, indicating a good agreement between

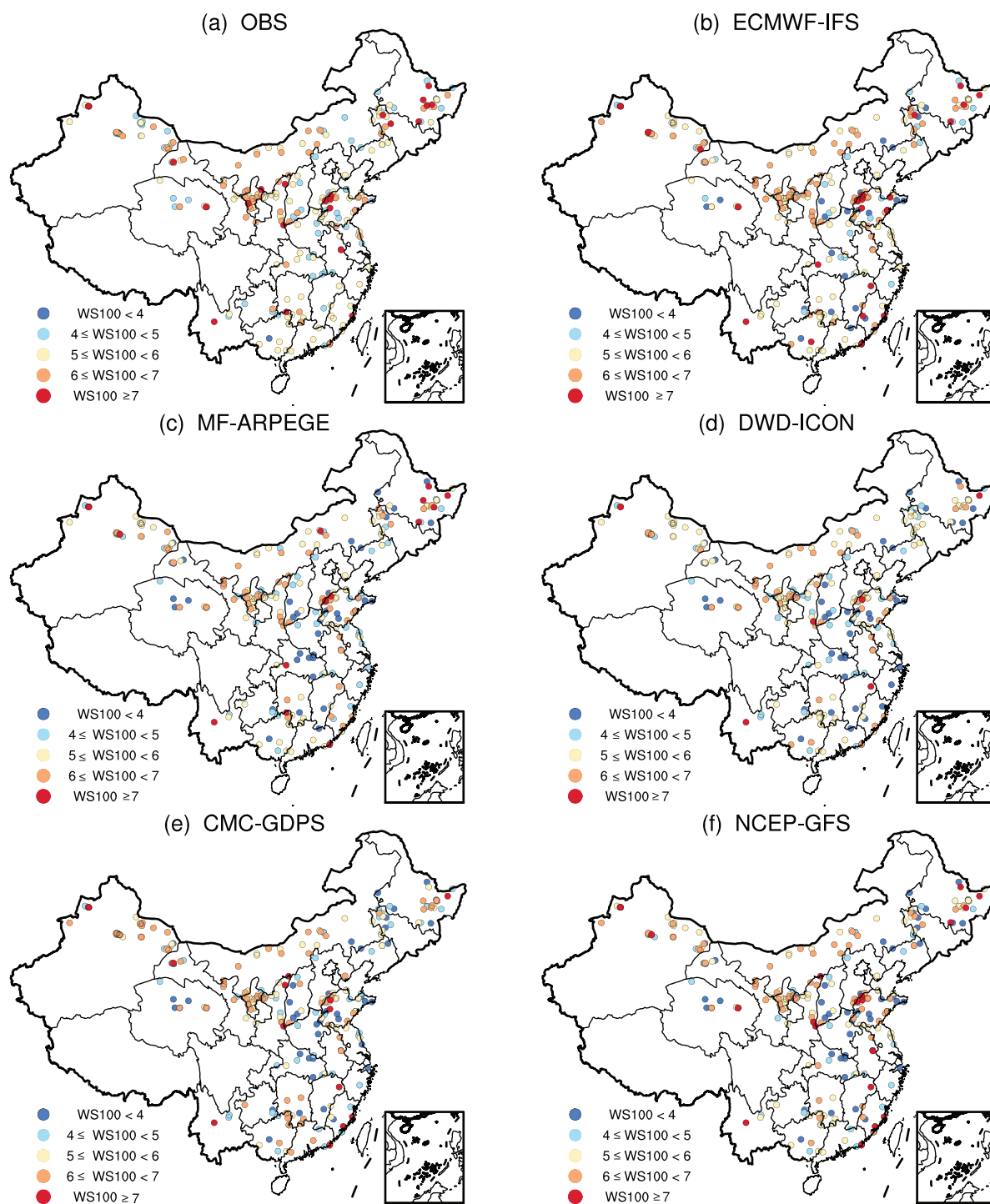


FIGURE 2 The spatial distribution of annual mean 100-m wind speed (unit: M s^{-1}) in (a) observation, (b) ECMWF-IFS, (c) MF-ARPEGE, (d) DWD-ICON, (e) CMC-GDPS and (f) NCEP-GFS.

models and observations. It was noted that DWD-ICON and CMC-GDPS displayed negative biases in northeast China, while CMC-GDPS showed a positive bias in southern China. The ECMWF-IFS model showed the lowest RMSE (2.34 m s^{-1}), the lowest MBE (-0.04 m s^{-1}), the highest TCC (0.74) and the highest PCC (0.58), making it the top-performing NWP model.

Following ECMWF-IFS, MF-ARPEGE, CMC-GDPS, DWD-ICON and NCEP-GFS ranked in decreasing order of performance in terms of RMSE and TCC (Table 1). However, CMC-GDPS exhibits the lowest PCC at 0.44, whereas DWD-ICON has highest MBE with -0.60 .

The probability density function (PDF) of wind speed at 100 m height is shown in Figure 3, where a bin size of

TABLE 1 The evaluation results of five NWP models for 100-m wind speed.

NWP model	PCC	TCC	RMSE	MBE
ECMWF-IFS	0.58	0.74	2.34	−0.04
MF-ARPEGE	0.54	0.72	2.45	−0.31
DWD-ICON	0.48	0.69	2.70	−0.60
CMC-GDPS	0.44	0.71	2.58	−0.37
NCEP-GFS	0.48	0.67	2.77	−0.34

0.5 m s^{-1} was used for both observations and NWP models. The observed PDF of wind speed at 100 m showed a skewed distribution with a peak around 4 m s^{-1} and a long tail at high wind speeds in China (Figure 3a). This pattern was reasonably replicated by the models, particularly the ECMWF-IFS. Among other models, DWD-ICON, NCEP-GFS, and CMC-GDPS overestimated (underestimated) the frequency of low (high) wind speed at $0\text{--}4 \text{ m s}^{-1}$ ($4\text{--}9 \text{ m s}^{-1}$), whereas MF-ARPEGE overestimated the frequency of middle wind speed between 2 and 5 m s^{-1} , which were shown similarly in characteristics of Northeast, Central, South and North China (Figure 3b–d,f). In Southwest China, the observed curve was reasonably simulated by MF-ARPEGE and CMC-GDPS, whereas ECMWF-IFS, DWD-ICON and NCEP-GFS showed some bias (Figure 3e). In Xinjiang, all five NWP models missed the observed PDF pattern, with the right (left) drift in ECMWF-IFS and MF-ARPEGE (DWD-ICON and NCEP-GFS) (Figure 3g).

The forecasting skills over six regions across China were further examined. As shown in Figure 4, the performance of forecasted 100-m wind speed varied in regions and models. The four models including ECMWF-IFS, MF-ARPEGE, CMC-GDPS and DWD-ICON consistently showed the higher accuracy of 100-m wind speed over Northeast and Southwest China, where the higher wind speed region locates. However, the model's ability in forecasting 100-m wind speed over Central and South China was relatively lower among all five models, indicating that the wind speed over Central and South China was more challenging for NWP models to predict than over other regions. The reason was that the wind speed in Central and Southern China was significantly influenced by a combination of weather systems originating from both the north (such as the East Asia trough, Arctic Oscillation) and the south (such as the subtropical high and typhoons), which was discussed in Li et al. (2023). It was noted that ECMWF-IFS consistently performed better than other models over all six regions, with the higher TCC and lower RMSE, while NCEP-GFS exhibited the poorest performance.

Figure 5 displays the distributions of the best-performing NWP model with highest TCC and lowest RMSE at each wind farm, which could provide helpful reference information for model users. It was shown that ECMWF-IFS performed the best generally, accounting for 63.3% and 64.8% of total 262 wind farms with the highest TCC and the lowest RMSE respectively. Notably, despite its relatively less frequent application in China's weather forecasting services, MF-ARPEGE demonstrated promising potential. It achieved 17.1% and 25.1% of wind farms with the best TCC and RMSE, respectively, among five NWP models, particularly over Northwest China. These results suggested that the adoption of MF-ARPEGE has benefits for wind energy forecasting in China. However, the remaining three models (CMC-GDPS, DWD-ICON and NCEP-GFS) only accounted for less than 20% in highest TCC and 10% in lowest RMSE of the total 262 wind farms.

4.2 | Seasonal and diurnal variations of wind speed

The observed monthly average of wind speed at 100 m AGL (represented by the black curve in Figure 6a) ranged from 4.9 to 6.4 m s^{-1} and exhibited double peaks during the year. The first peak of wind speed at 100 m AGL occurred during February to May, reaching the values of 6.1 to 6.4 m s^{-1} before sharply declining to around 5 m s^{-1} in August and September. The second peak occurs in November with a speed of 6 m s^{-1} , followed by a decrease to less than 5 m s^{-1} in January. All the NWP models were able to capture the seasonal variations of the wind speed at 100 m, while exhibiting a negative bias throughout the year (Figure 6a). Among five NWP models, ECMWF-IFS outperformed other models with an averaged bias of -0.05 m s^{-1} , while DWD-ICON underperformed other models with the averaged bias up to -0.6 m s^{-1} .

In terms of the diurnal cycle of wind speed at 100 m (Figure 6b), the observed hourly averaged wind speed exhibited a “smile curve” shape, with the lowest wind speed occurring at 11:00 BJT. The wind speed increased during the afternoon (12:00–20:00 BJT), remaining high values throughout the night until early morning the next day. Three NWP models (ECMWF-IFS, MF-ARPEGE and CMC-GDPS) were able to capture the characteristics of the observed “smile curve”, although their lowest point occurred at 12:00 BJT, with a lag of an hour compared with the observation (Figure 6b). The forecasts from ECMWF-IFS (red curve in Figure 6b) matched the observation during the night period (20:00–11:00 BJT) closely but exhibited a negative bias during afternoon. On the

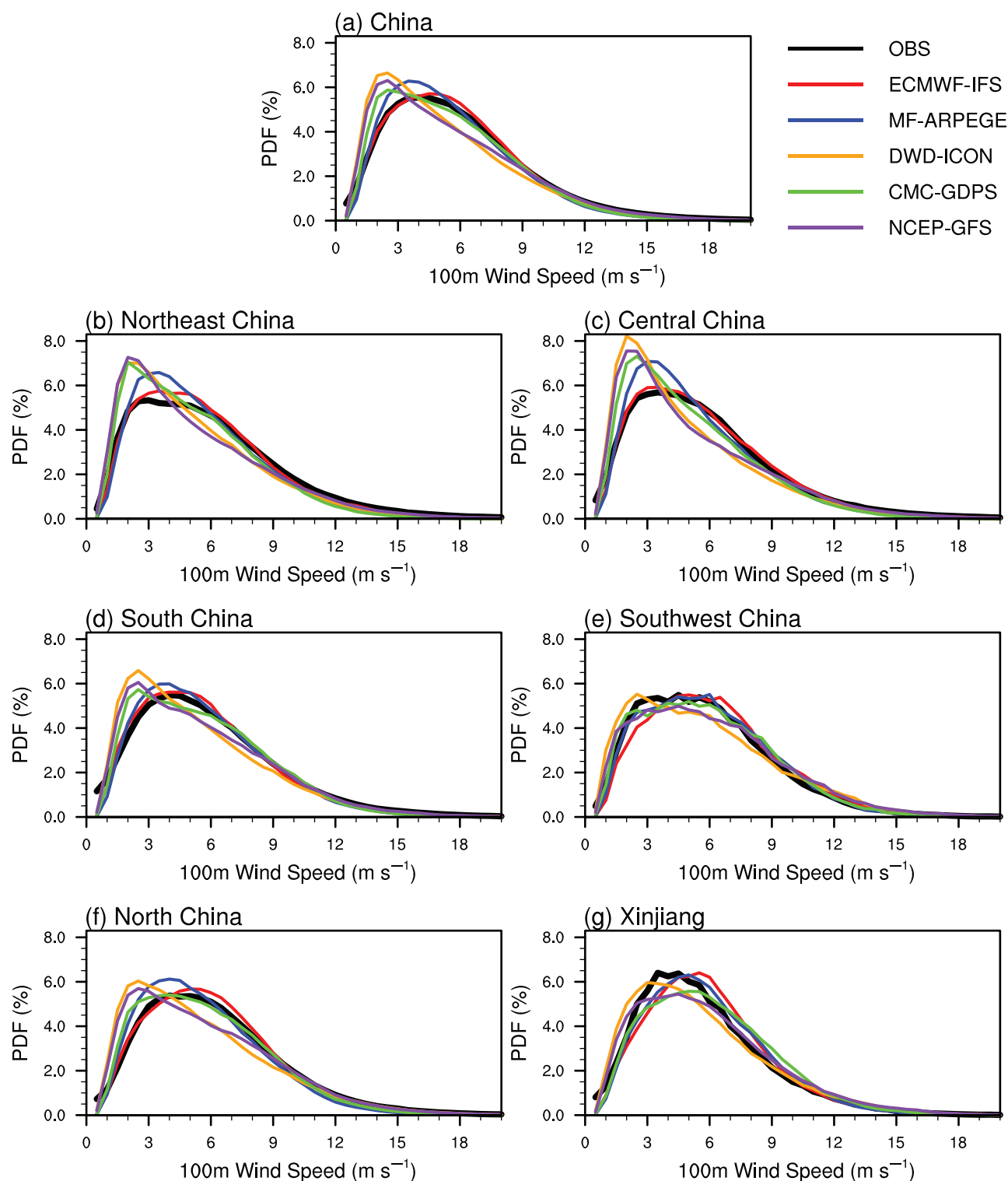


FIGURE 3 Observed (thick black) and forecasted (thin coloured) probability distribution function (unit: %) of 100-m wind speed in (a) China and six regions: (b) Northeast China, (c) Central China, (d) South China, (e) Southwest China, (f) North China and (g) Xinjiang.

contrary, the forecasts from NCEP-GFS performed well during the afternoon while showed a large negative bias during nighttime to morning.

To quantitatively illustrate the models' performance in capturing the seasonal and diurnal variation of 100-m

wind speed, the heatmap of each model depicting the variations of RMSE across months and hours was presented in Figure 7. Seasonally, the high RMSE of wind speed occurred in spring and peaked in May, which was consistent with the climatological high wind speed in

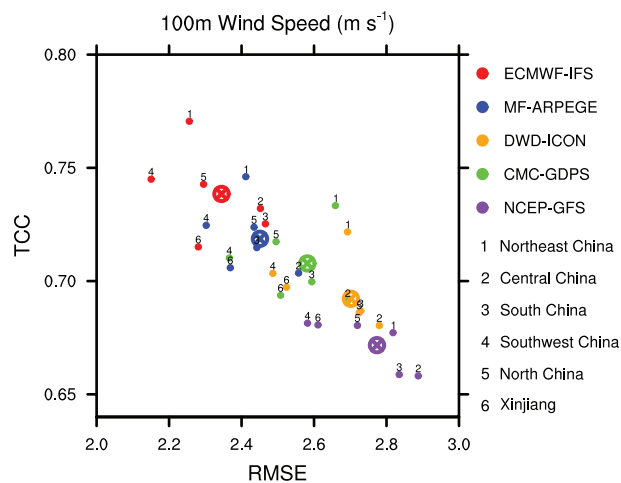


FIGURE 4 The TCC (unit: 1) and RMSE (unit: m s^{-1}) of five NWP models for 100-m wind speed in six regions (solid dots) and the average of China (hollow circle with an X).

spring (Figure 6a). It was observed that all the five NWP models exhibited a consistent pattern, showing higher (lower) RMSE during nighttime (noon). The diurnal distribution of RMSE can be attributed to the influence of boundary layer stability and nocturnal conditions. Nocturnal stabilization can lead to decoupling between the surface layer and the atmospheric boundary layer above, limiting vertical mixing. As a result, NWP models may struggle to accurately simulate these stable conditions, resulting in larger wind speed biases and RMSE during the nighttime (Heppelmann et al., 2017). By examining each NWP model individually, ECMWF-IFS demonstrated the lowest RMSE values both diurnally and seasonally, indicating its better performance in predicting the variations of 100-m wind speed, while NCEP-GFS exhibited the largest RMSE values.

4.3 | Attributing analysis of models errors in wind speed

To gain further understanding of the ability and the errors of NWP models, the error decomposition method was employed to diagnose the sources of error in 100-m wind speed forecasts within NWP models. The averaged RMSE of all wind farms could be decomposed into three components: the bias component, sequence error component and distribution error component. As shown in Figure 8, SEQU was the primary source of model error in forecasting hub-height wind speed, contributing $\sim 70\%$ of the total error. On the other hand, BIAS and DIST accounted for a relatively smaller proportion (approximately 20% and 10%, respectively) of the total error,

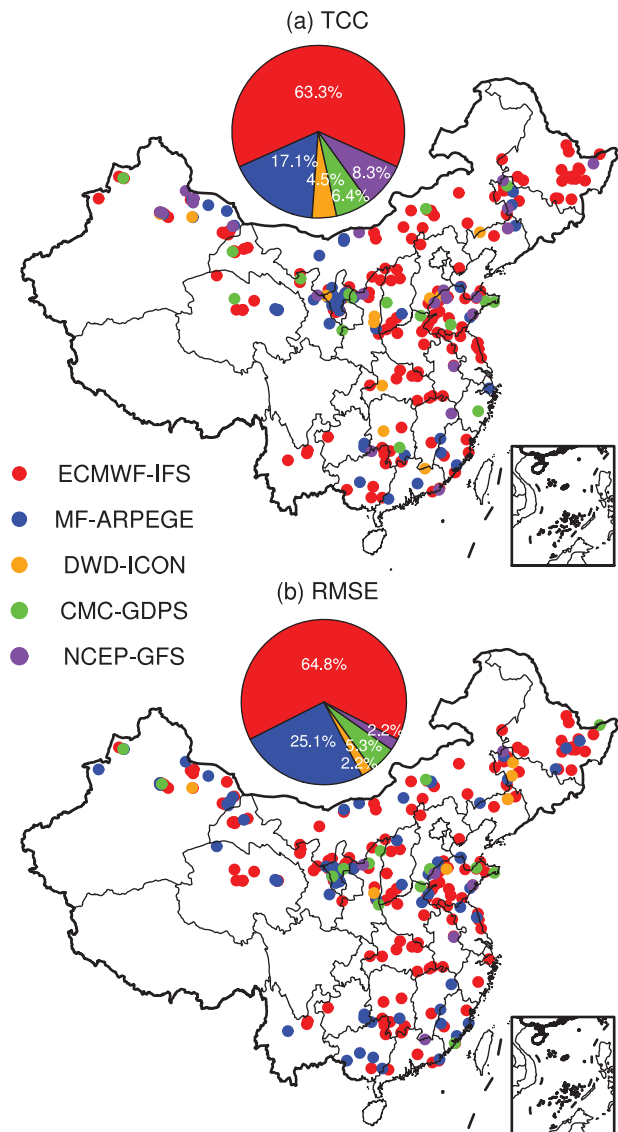


FIGURE 5 Distribution of (a) TCC, (b) RMSE for best performance NWP model and pie chart percentage. Red, blue, yellow, green and purple point denotes ECMWF-IFS, MF-ARPEGE, DWD-ICON, CMC-GDPS and NCEP-GFS is the best performance NWP model, respectively. The pie chart at the top represents the percentage of the best performance NWP model.

although their maximum values were comparable to SEQU errors. This implied that the forecast errors in NWP models were mainly caused by the temporal inconsistency between model and observation, that is, being ahead of or lagging the observed data. It was noted that the median values of SEQU were higher than their average, indicating that most wind farms showed relatively huge SEQU errors. This reflects the challenges of global NWP models in accurately forecasting the timing of wind speed variations. On the contrary, the median values of BIAS and DIST were both lower than their average and

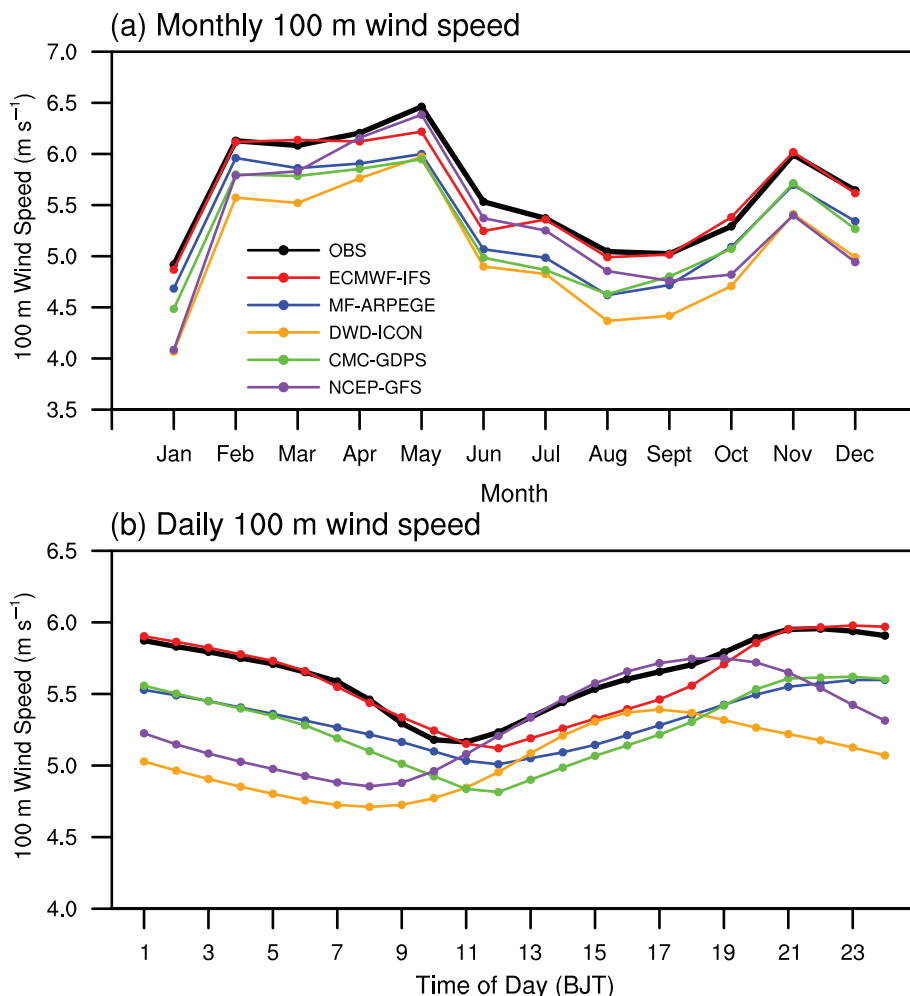


FIGURE 6 (a) Seasonal and (b) diurnal variation of observed (thick black) and forecasted (thin coloured) mean wind speed at 100 m (units: m s^{-1}).

considerably smaller than the maximum values, indicating that the majority of wind farms show small BIAS and DIST errors except for a few stations. ECMWF-IFS displayed the lowest values consistently in BIAS, SEQU and DIST components among five models. In comparison with ECMWF-IFS, MF-ARPEGE displayed slightly lower error of SEQU, while with higher error of BIAS and DIST.

Figure 9 shows the diurnal variation of RMSE for 100-m wind speed and its decomposed components in NWP models, on bases of the annual mean (Figure 9a–d) and seasonal mean (Figure 9e–t) respectively. It was observed in Figure 9a that all the five models displayed lower (higher) RMSE during daytime (nighttime). Consistent with the mean state, the SEQU was the main contributor to the total RMSE of forecasted wind speed (Figure 9b–d). The SEQU increased with the lead time of forecasts, showing the higher value after 12:00 BJT with the maximum reaching 2.0 m s^{-1} (Figure 9c). However, although SEQU dominated the magnitude of wind speed's RMSE, the diurnal variation of forecasted wind speed's RMSE was largely determined by the BIAS and

DIST. Among five models, ECMWF-IFS showed the lowest diurnal BIAS and DIST, leading to the lowest total RMSE throughout the day. Despite the higher BIAS, the NCEP-GFS outperformed other models with the lowest SEQU error before 7:00, indicating its advantage in predicting the temporal variation of wind speed during early hours of the forecasts. Seasonally, the daily variations of total RMSE and its components are generally consistent with those of annual mean. Meanwhile, it is noted that the SEQU during cold seasons (winter and fall) is consistently lower than that in warm seasons (summer and spring) in all the models, which indicates that the forecasted errors due to the mismatched timing between predication and observation is relatively lower in winter seasons.

4.4 | Wind ramp events

In this section, the performance of simulated ramp events in NWP models were evaluated. A ramp-specific metric was used to provide an appropriate measure of model

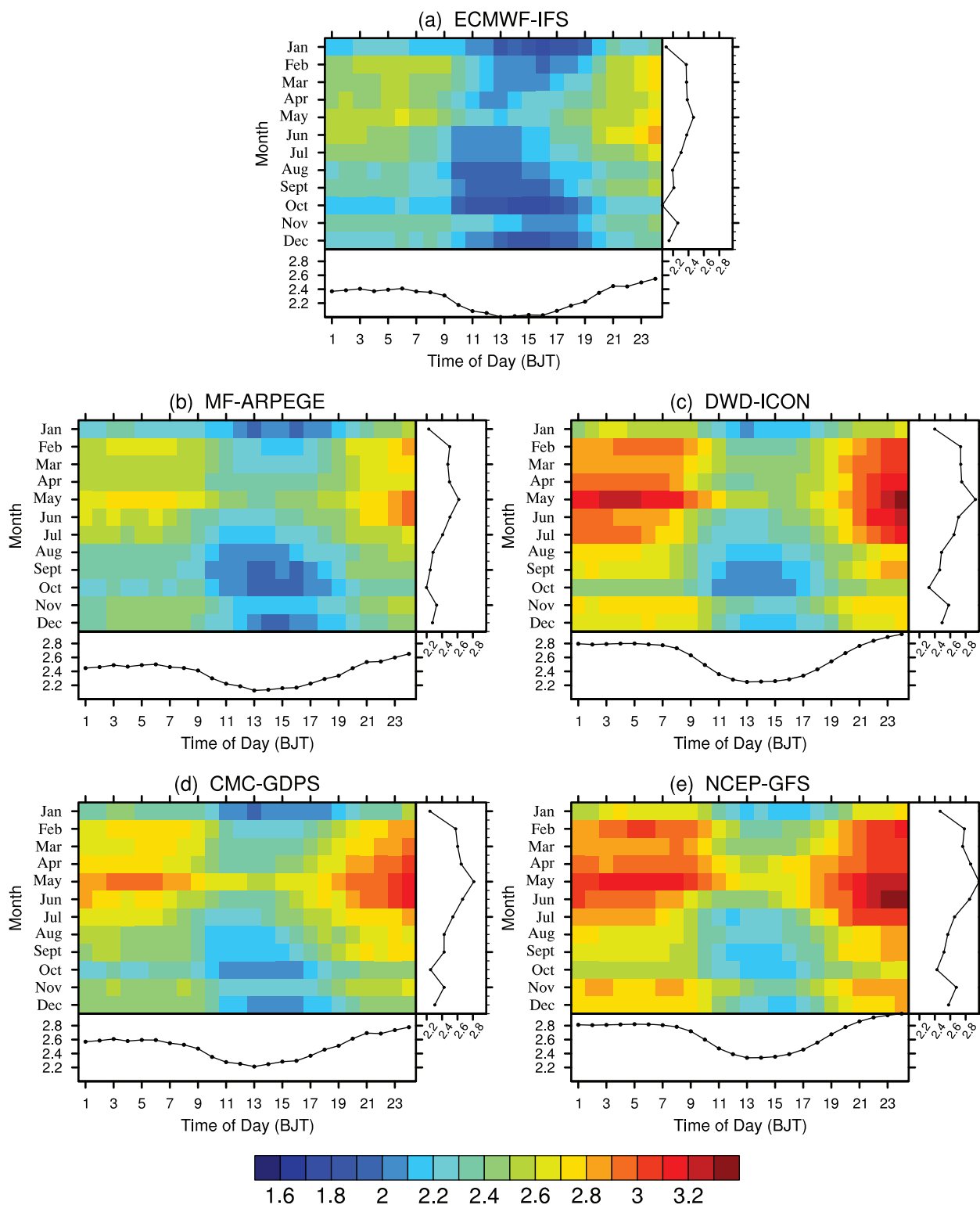
RMSE of 100 m Wind Speed (m s^{-1})

FIGURE 7 Seasonal and diurnal variation (shadings) of RMSE of wind speed at 100 m (unit: m s^{-1}) for (a) ECMWF-IFS, (b) MF-ARPEGE, (c) DWD-ICON, (d) CMC-GDPS and (e) NCEP-GFS. Black curves in right and bottom panel denote the daily and annual mean.

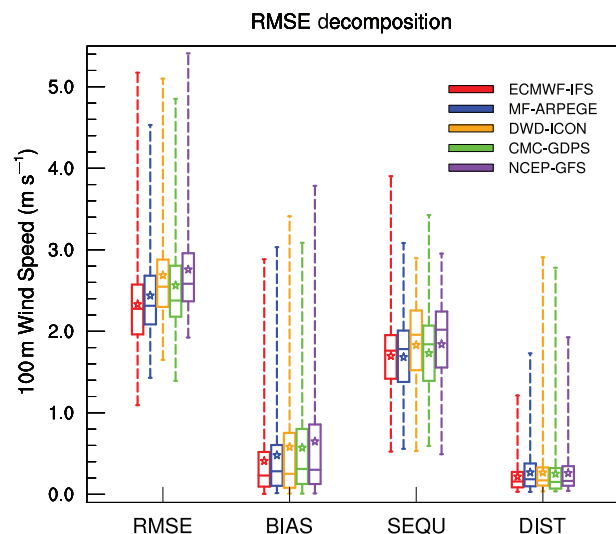


FIGURE 8 Box plot of mean RMSE and decomposed BIAS, SEQU and DIST components for 100-m wind speed (unit: M s^{-1}) derived from ECMWF-IFS, MF-ARPEGE, DWD-ICON, CMC-GDPS and NCEP-GFS. The line from top to bottom represents maximum, 75%, median, 25%, and minimum, respectively. The stars denote mean value averaged over all wind farms.

skill. Figure 10 displays the number of wind ramp events in observation and simulations detected using the min-max identification method, which considered multiple ramp power thresholds and duration as mentioned in Section 3. It was shown that there were numerous small-amplitude and long-duration ramps (bottom-right corner of each panel) while relatively few occurrences of extreme ramp events (top-left corner of each panel) in the observation (Figure 10a). The distributions of wind ramp events derived from NWP models resembled that from observation generally (Figure 10b–f). However, the number of wind ramp events in NWP models was consistently less than those in observations at all the threshold of wind ramps. For example, the overserved number of ramp events with 50% power change within 120 min exceeded 300 times during a year, which was dramatically underestimated in models and showing less than 40 and 100 times in ECMWF-IFS and NECP-GFS, respectively. This discrepancy can be attributed to the fact that the models exhibited the smoother wind fields compared with the observed temporal series of power at a spatial resolution of 12–25-km grid spacing and temporal model output frequency of 1–3 h. This smoothing effect tended to reduce the occurrence of ramp events in the NWP model outputs, which was also discussed in Bianco et al. (2016) using 13 km-horizontal grid spacing model and tall-tower anemometer observations. Among five NWP models, NCEP-GFS and DWD-ICON detected relatively

more wind ramp events compared with other models due to their higher time resolution output (1 h) than other models (3 h), which helped to capture more rapid changes in wind speed and wind power.

Figure 11 presents the skill score matrices for the simulations of five NWP models using the min-max ramp definition method. The results showed that, the skill scores were generally lower for ramp event with the duration of 60 min than those of ramp events with the duration of 120–180 min, particularly for huge changes in power of more than 50%, which was also noted in Ouyang et al. (2017). This indicated that it was more difficult for models to identify and capture the characteristics associated with extreme wind ramps during short time. NCEP-GFS still exhibited higher skill scores than other models in the forecast of wind ramps, particularly for those during 120 and 180 min, with the skill score reaching 0.1. It was reported that the skill score of model output can reach 0.3 with 15-min intervals (Bianco et al., 2016). Therefore, it is of significant importance to improve the model output frequency for forecasting wind ramp events.

5 | CONCLUSION AND DISCUSSION

The forecasting performance of wind speed at turbine hub height in China was evaluated across five NWP models against observations from 262 wind farms. The ability of NWP models to forecast the climatological features of wind speed at 100 m over China was examined. The assessment suggested that the spatial distribution of climatological wind speed at 100 m was reasonably forecasted by the NWP models, showing the higher wind speed over Northern China. The forecasting skills over six regions across China indicated that the accuracy of 100-m wind speed over Northeast and Southwest was higher than that over Central and South China. Among five NWP models, ECMWF-IFS showed the highest skill in predicting the mean state of 100-m wind speed, with the highest TCC/PCC (0.74/0.58) and lowest RMSE/MBE ($2.34/-0.04 \text{ m s}^{-1}$). By examining each wind farm, ECMWF-IFS performed the best generally, accounting for 63.3% and 64.8% of total 262 wind farms with the highest TCC and the lowest RMSE among all five models, respectively, followed by MF-ARPEGE with 17.1% and 25.1% of wind farms for the best TCC and RMSE.

The seasonal variations of wind speed at 100 m were reasonably forecasted by all the models, showing two hub-height wind speed peaks, occurring in May (6.4 m s^{-1}) and November ($\sim 6.0 \text{ m s}^{-1}$), while the

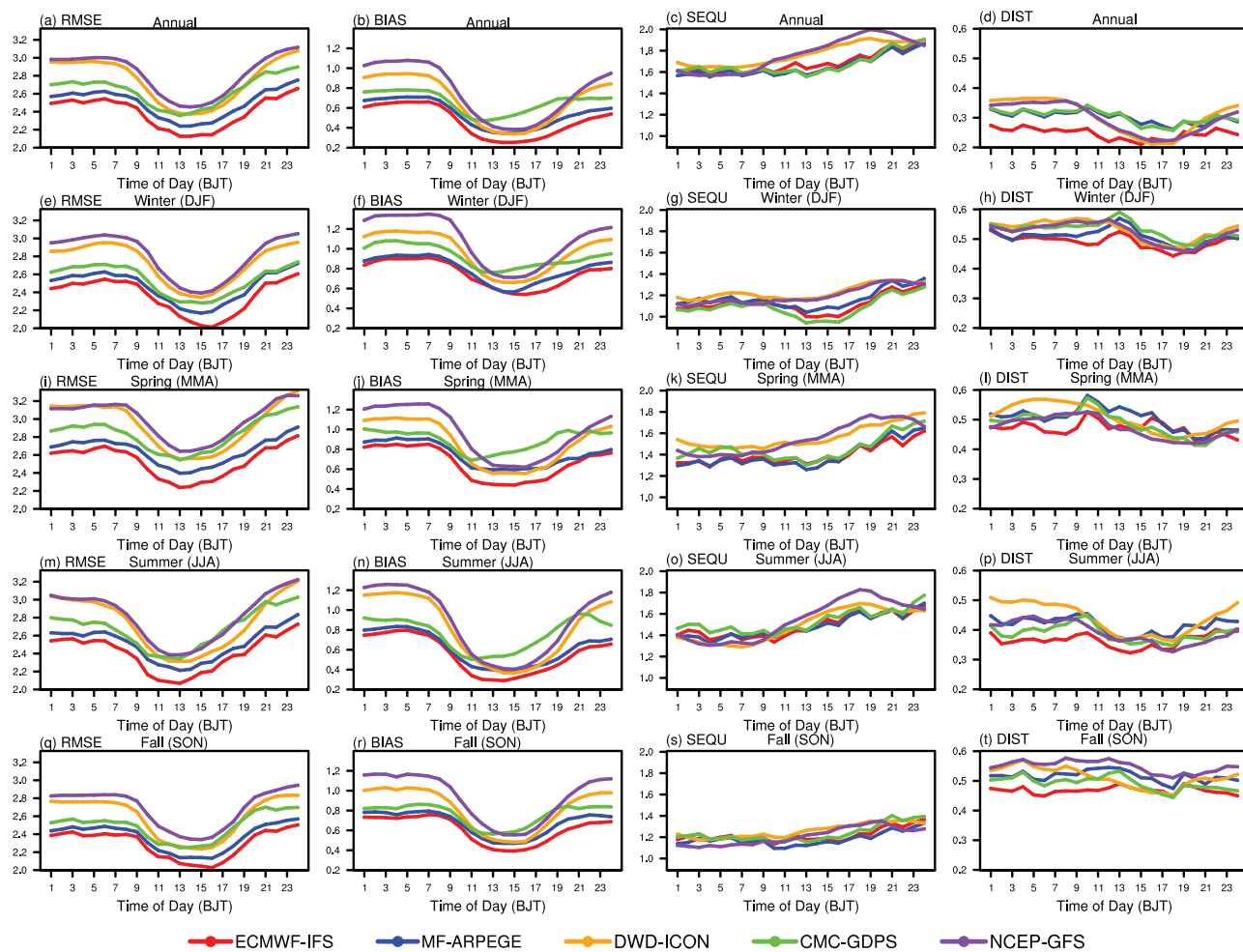
Daily RMSE Decomposition of 100m Wind Speed (m s^{-1})

FIGURE 9 The annual-averaged diurnal variation of (a) RMSE, decomposed (b) BIAS, (c) SEQU and (d) DIST components for 100-m wind speed (unit: M s^{-1}) derived from ECMWF-IFS, MF-ARPEGE, DWD-ICON, CMC-GDPS and NCEP-GFS. (e–h), (i–l), (m–p) and (q–t) are same with (a–d), except for winter (DJF), spring (MMA), summer (JJA) and fall (SON).

magnitude was underestimated throughout the year by -0.05 m s^{-1} (ECMWF-IFS) to -0.6 m s^{-1} (DWD-ICON). In terms of diurnal variation, only three NWP models (ECMWF-IFS, MF-ARPEGE and CMC-GDPS) were able to capture the characteristics of the observed “smile curve”, showing the high wind speed during nighttime despite the 1-hour lag of lowest wind speed compared with the observations. It was observed that all the five NWP models exhibited a consistent pattern, showing higher RMSE during nighttime, indicating that it was more difficult to accurately simulate the wind speed at 100 m under nocturnal stable conditions. ECMWF-IFS displayed the lowest RMSE values both diurnally and seasonally, while NCEP-GFS exhibited the largest RMSE values.

The further error decomposition analysis revealed that the error in 100-m wind speed forecasts within the

NWP models was primarily attributed to the temporal sequence error component, which accounted for $\sim 70\%$ of the total error. This implied that the 100-m wind speed forecast errors in NWP models were mainly caused by the temporal inconsistency between model and observation. It was noted that the median values of SEQU were higher than its average, indicating that the models show relatively large errors at most wind farms. BIAS and DIST only accounted for a relatively smaller proportion ($\sim 20\%$ and 10% respectively) of the total error, although their maximum values were comparable with SEQU errors.

The assessment of simulated wind ramp events showed that the number of wind ramp events was significantly underestimated by NWP models, which may be attributed to the low spatial and time resolution of models. However, NCEP-GFS and DWD-ICON

Wind ramp events

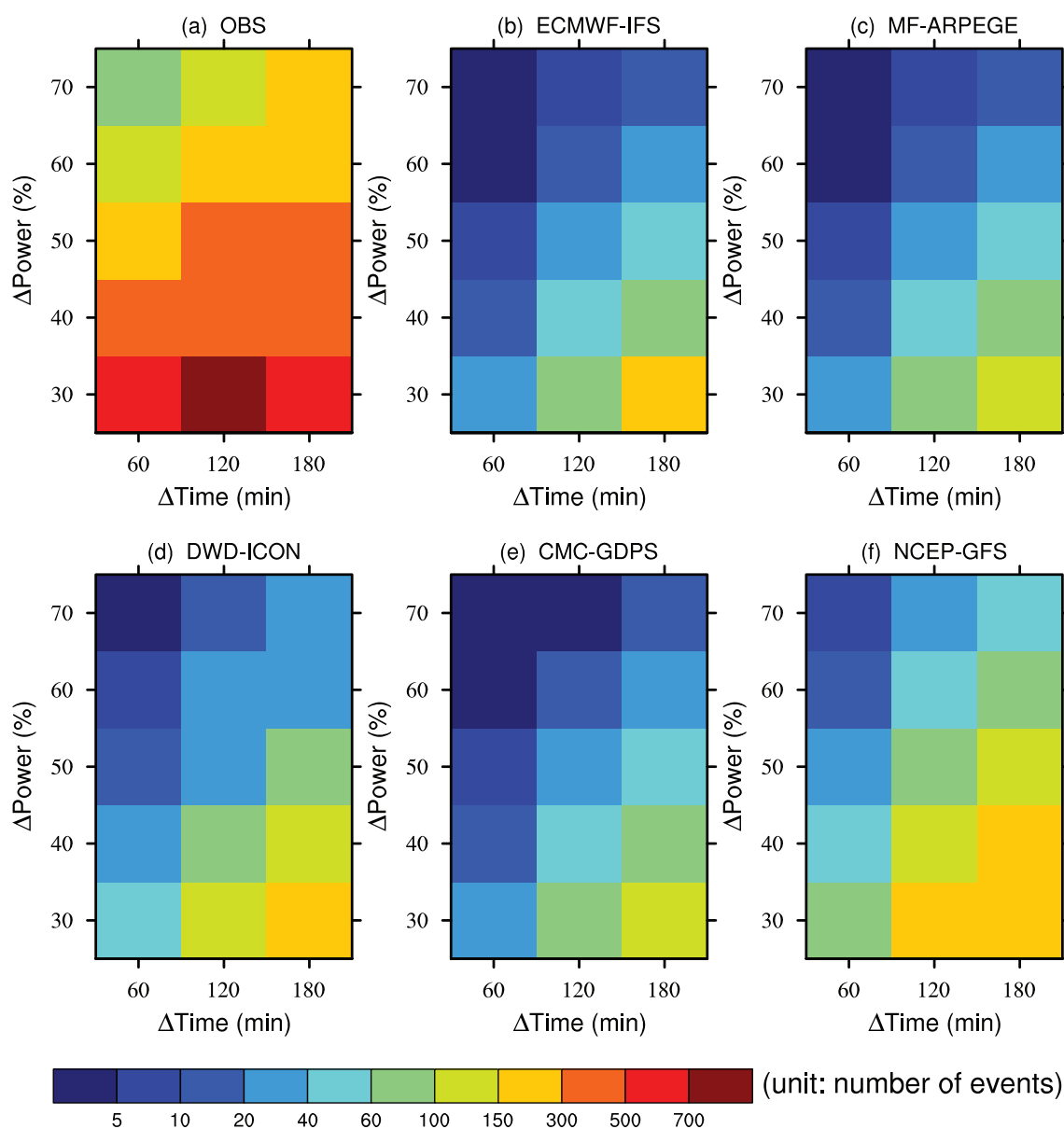


FIGURE 10 Number of wind ramp events derived from (a) observation, (b) ECMWF-IFS, (c) MF-ARPEGE, (d) DWD-ICON, (e) CMC-GDPS and (f) NCEP-GFS. The power threshold of ramp definition ranges from 30% to 70%, and the window length ranges from 60 to 180 min.

performed relatively better due to their higher time resolution, with the skill score of wind ramps during 180 min reaching 0.1. Therefore, it would be worthwhile to increase the model resolution to 15-min intervals to improve the performance of wind ramp events.

The current study provided a systematic assessment of five NWP models' ability to predict the climatology, seasonal and diurnal variations of turbine hub-height wind speed in China utilizing observation from 262 wind

farms. These results helped to better understand the ability of NWP models to predict hub-height wind speed over China, providing valuable information for further investigation and improvement. Apart from global NWP models, other approaches, such as using rapidly updated high-resolution regional models (Bianco et al., 2016; Macleod et al., 2018), multi-model ensemble (Doan et al., 2019) and machine learning algorithms (Moss et al., 2023), have been examined showing promising

Wind Ramp Skill Score

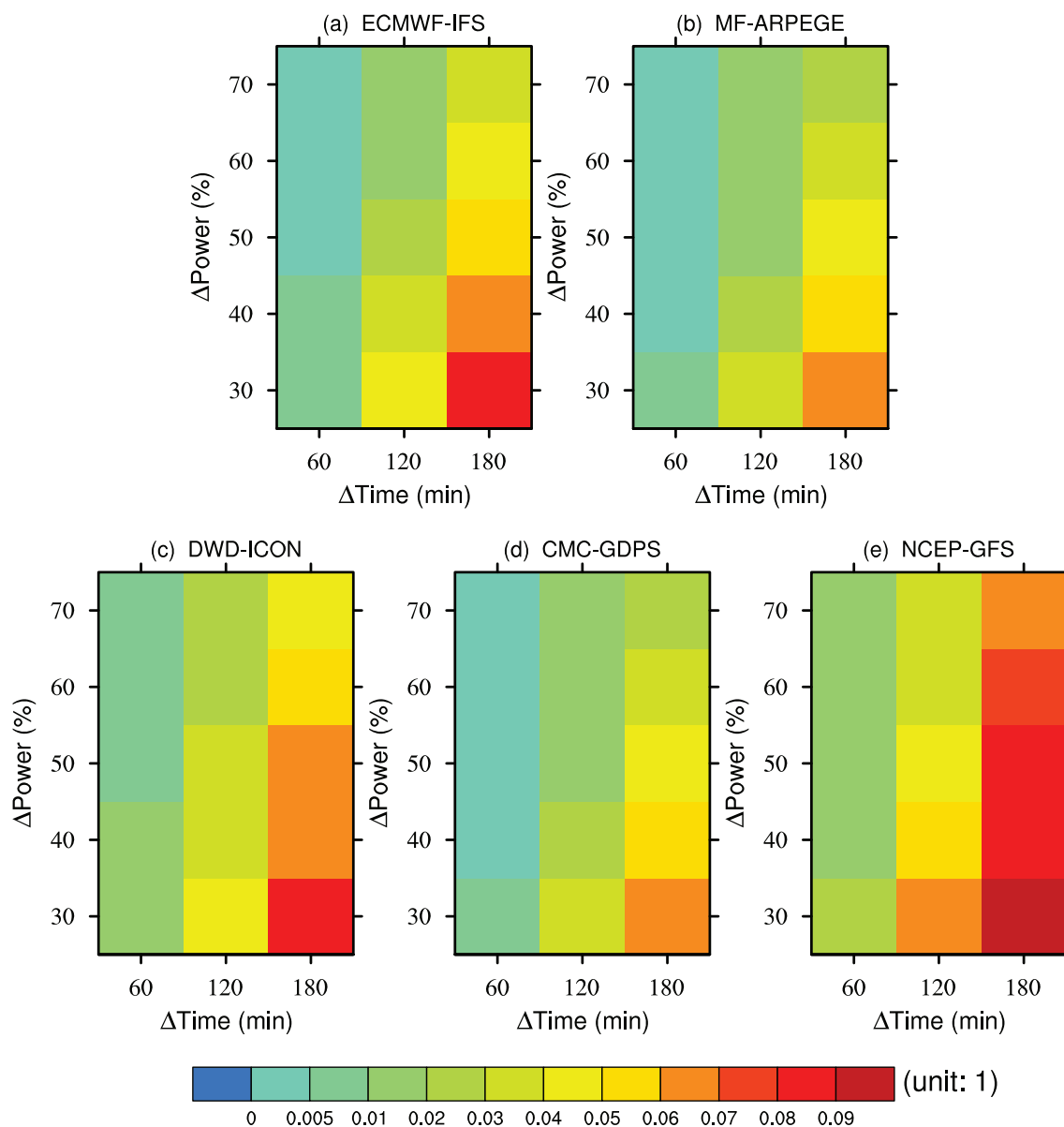


FIGURE 11 Matrix of wind ramp skill scores (unit: 1) derived from (a) ECMWF-IFS, (b) MF-ARPEGE, (c) DWD-ICON, (d) CMC-GDPS and (e) NCEP-GFS.

performances. These approaches will be subject to further examination in future research efforts.

AUTHOR CONTRIBUTIONS

Chenxi Jin: Visualization; writing – original draft; conceptualization; methodology; investigation. **Yang Yang:** Conceptualization; methodology; writing – review and editing; validation. **Chao Han:** Resources; writing – review and editing. **Ting Lei:** Writing – review and editing. **Chen Li:** Writing – review and editing. **Bing Lu:** Writing – review and editing.

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ARPEGE, DWD-ICON, CMC-GDPS and NCEP-GFS teams for their efforts in making the data available.

DATA AVAILABILITY STATEMENT

The authors do not have permission to share the observed data. All NWP models output data were available in their website as below (useful before January 2024): ECMWF-IFS data: <https://www.ecmwf.int/en/forecasts/datasets/set-i>. MF-ARPEGE data: <https://mf-models-on-aws.org/en/doc/tools/partial-download/>. DWD-ICON data: <https://opendata.dwd.de/weather/nwp/>. CMC-GDPS data: https://dd.meteo.gc.ca/model_gem_global/. NCEP-GFS data: <https://nomads.ncep.noaa.gov/pub/data/nccf/com/gfs/prod/>.

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